D. API & Contract Governance

Standalone capability brief · Suggested module: **API Standards + Contract Platform**

Purpose

Make APIs and events predictable, discoverable, testable, and evolvable across product teams and external consumers.

1. Scope and capability areas

Schema registry

Store and discover API, event, and message schemas with ownership, versions, and compatibility rules.

Contract-first development

Support specification-first design, validation, code generation, and review before implementation.

Consumer-driven testing

Verify provider changes against known consumer expectations and important integration contracts.

Lifecycle policies

Define deprecation notices, sunset dates, migration guidance, and compatibility support windows.

Common conventions

Standardize pagination, filtering, sorting, bulk operations, error formats, and stable error codes.

SDK generation

Generate and publish client SDKs with explicit language support, release versions, and compatibility guarantees.

2. Shared contracts and standards

- Every externally consumed API has an owner, documented schema, authentication model, and lifecycle state.
- Compatibility rules distinguish additive changes from breaking changes and cover both request and response behavior.
- Errors use a common envelope with stable machine-readable codes and safe human-readable messages.
- Deprecation includes a timeline, telemetry, migration documentation, and a communication plan.

3. Recommended implementation approach

- Run schema linting and compatibility checks in CI before merge or release.
- Publish API and event definitions to a searchable catalog.
- Generate clients from the approved schema rather than from ad hoc implementation details.
- Use consumer-driven contract tests for high-risk integrations.
- Track API usage to identify consumers before removal.

4. Failure behavior and operational expectations

- If schema publication fails, do not silently ship an undocumented breaking contract.
- If a downstream consumer is incompatible, block release or require an explicit reviewed exception.

- Unknown error codes should degrade to a safe generic message while preserving diagnostic details for operators.

5. Suggested operational metrics

- Breaking changes detected before release
- Contract-test pass rate
- Deprecated API usage
- Time to publish SDKs after API release
- Consumer incidents caused by contract drift

6. Readiness and acceptance checklist

- ■ New APIs meet documented conventions and are discoverable.
- ■ Compatibility checks run automatically in CI.
- ■ Deprecation and sunset procedures are repeatable.
- ■ Generated SDKs map consistently to approved API schemas.

Ownership note: assign a named team, service owner, support route, versioning policy, and deprecation process before treating this capability as a platform contract.